

Physical Data Model

After discussing and analyzing, we choose :

- DBMS: SQLite
- Data Model Type: Relational

1. Product

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		productID	Integer	Yes	ID, auto increment
2			title	VARCHAR	Yes	Name of the product
3			category	VARCHAR	Yes	Product category (Book, CD, DVD, or LP record)
4			value	FLOAT	Yes	Base price of the product before VAT
5			price	FLOAT	Yes	Selling price of the product
6			quantity	Integer	Yes	Available quantity in stock
7			description	VARCHAR	No	Product description
8			imageUrl	VARCHAR	No	The URL for the image of the product item
9			barCode	VARCHAR	No	Unique barcode for the product
10			warehouseEntryDate	DATE	No	Date when the product entered the warehouse
11			dimensions	VARCHAR	No	Physical dimensions (e.g., 10x20x5 cm)
12			weight	FLOAT	No	Weight in kilograms
13			warehouseProvince	VARCHAR	No	Province of warehouse storing the product
14			warehouseDistrict	VARCHAR	No	District of warehouse storing the product
15			warehouseAddress	VARCHAR	No	Detailed address of warehouse storing the product

2. CD

#	PK	FK	Column Name	Data type	Mandatory	Description
1		x	productID	INTEGER	Yes	ID, same as ID of Product of which type is CD
2			artist	VARCHAR(100)	Yes	Name of the artist or band
3			recordLabel	VARCHAR(100)	Yes	Name of the record label
4			tracklist	VARCHAR(1000)	No	List of tracks in the CD
5			genre	VARCHAR(100)	Yes	Music genre of the CD
6			releaseDate	DATE	Yes	Official release date of the CD

3. Book

#	PK	FK	Column Name	Data type	Mandatory	Description
1		x	productID	INTEGER	Yes	ID, same as ID of Product of which type is Book
2			author	VARCHAR(100)	Yes	Author of the book
3			coverType	VARCHAR(45)	Yes	Type of cover (paperback or hardcover)
4			publisher	VARCHAR(100)	Yes	Publisher of the book
5			publicationDate	DATE	Yes	Date of publication
6			pages	INTEGER	Yes	Number of pages
7			language	VARCHAR(45)	Yes	Language of the book
8			genre	VARCHAR(100)	Yes	Genre of the book

4. DVD

#	PK	FK	Column Name	Data type	Mandatory	Description
1		x	productID	INTEGER	Yes	ID, same as ID of Product of which type is DVD
2			discType	VARCHAR(45)	Yes	Type of disc (e.g: Blu-ray, HD-DVD)
3			director	VARCHAR(100)	Yes	Name of the artist
4			runtime	VARCHAR(20)	Yes	Total runtime of the DVD
5			studio	VARCHAR(100)	Yes	Name of the studio
6			language	VARCHAR(45)	Yes	Language of the content in DVD
7			subtitle	VARCHAR(45)	Yes	Language subtitle
8			releaseDate	DATE	Yes	Release date of the DVD
9			genre	VARCHAR(100)	Yes	The genre of the DVD

5. DeliveryInfo

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		deliveryinfoID	INTEGER	Yes	ID, auto increment
2			recipientName	VARCHAR(100)	Yes	Name of the recipient
3			email	VARCHAR(100)	Yes	Email address
4			phoneNumber	VARCHAR(20)	Yes	Phone number
5			province	VARCHAR(45)	Yes	Province/city
6			district	VARCHAR(45)	Yes	District
7			address	VARCHAR(200)	Yes	Delivery address
8			deliveryMethod	VARCHAR(20)	Yes	Delivery method (Regular/Rush)
9			deliveryTime	FLOAT	No	Requested time for rush delivery
10			isRushDeliveryEligible	BOOLEAN	Yes	Indicates if address is eligible for rush delivery

11			deliveryInstructions	VARCHAR(255)	No	Special delivery instructions
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6. Invoice

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		invoiceID	INTEGER	Yes	Unique Identifier for the invoice
2		x	orderID	INTEGER	Yes	Order associated with this invoice (FK to Order)
3			issueDate	DATE	Yes	Date when invoice was created
4			subtotal	FLOAT	Yes	Total price excluding VAT
5			VAT	FLOAT	Yes	Value Added Tax amount (10% of subtotal)
6			totalWithVAT	FLOAT	Yes	Subtotal + VAT
7			regularDeliveryFee	FLOAT	Yes	Fee for regular delivery items
8			rushDeliveryFee	FLOAT	Yes	Fee for rush delivery items
9			totalAmount	FLOAT	Yes	Final amount to be paid
10			paymentMethod	VARCHAR(45)	Yes	Method of payment
11			VAT_RATE	FLOAT	Yes	Value-added tax rate (constant 10%)

7. Payment Transaction

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		transactionID	INTEGER	Yes	Unique Identifier for the transaction
2			amount	FLOAT	Yes	Payment amount
3			paymentMethod	VARCHAR(45)	Yes	Method of payment
4			transactionDate	DATE	Yes	Date of transaction
5			status	VARCHAR(20)	Yes	Transaction status (Pending, Success, Failed...)
6			content	VARCHAR(255)	No	Transaction details
7		x	cardID	INTEGER	Yes	ID of used card
8		x	invoiceID	INTEGER	Yes	FK to Invoice

8. User

#	PK	FK	Column Name	Data type	Mandatory	Description
1	x		userID	INTEGER	Yes	Randomly auto-generated, exclusive
2			username	VARCHAR(50)	Yes	Exclusively exists to identify users using the software
3			email	VARCHAR(100)	No	Saved when using Gmail for registration
4			password	VARCHAR(255)	Yes	Must be hashed before stored into database
5			role	VARCHAR(45)	Yes	User role (Administrator / Product Manager)
6			status	BOOLEAN	Yes	True = Unblocked, False = Blocked

